

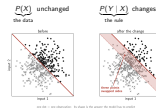


Good afternoon. Our subject is the blind spot paradox: a model that repairs itself can destroy the very evidence its own monitor needs. In the next twenty minutes we will present what we formalized, what we measured, and the single reading that ties the two together.

The Blind Spot Paradox

└ The problem

└ Concept drift: the rule changes, the data does not



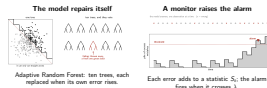
A model trained before the change is now **wrong** on the shaded band.

A classifier learns a rule from data. In a data stream, that rule can change while the inputs keep exactly the same distribution. This is concept drift: P of X does not move, P of Y given X does. It is the normal situation for a financial signal, where the same inputs stop meaning the same thing. In the bench we use, points in the plane are separated by a line. At the change, the line moves. Every point in the shaded band switches label, so a model trained before the change keeps predicting confidently, and is now wrong on that band. The size of the band is the size of the drift, written δ .

The Blind Spot Paradox

└ The problem

└ Two defences that run at the same time

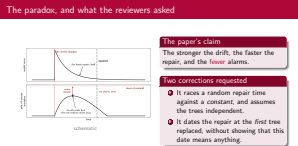


Two defences are deployed together. The first one lives inside the model. The Adaptive Random Forest is made of ten decision trees that vote. Each tree watches its own error with a small detector, and when that error rises, the tree is replaced by a fresh one. The forest repairs itself locally, without asking anyone. The second defence sits outside. A cumulative detector, a Page-Hinkley test, watches the error of the whole forest. Every error pushes a statistic up by almost one, every correct answer pulls it down a little, never below zero. When the statistic crosses a threshold λ , the alarm fires, and that alarm is the only thing a human operator sees.

The Blind Spot Paradox

└ The problem

└ The paradox, and what the reviewers asked



When both defences run together, something unexpected happens. The drift starts, errors pile up in the detector, but the forest replaces trees, the errors stop, and the pile drains before it reaches the threshold. The paper we study calls this the blind spot paradox: the stronger the drift, the faster the forest repairs itself, and the fewer alarms the operator gets. The paper was reviewed, and two corrections were requested. First, it compares a random repair time with a constant detection time, and it assumes the ten trees are independent without proving it. Second, it dates the repair at the moment the first tree is replaced, without showing that this moment has anything to do with the error disappearing. These two corrections are where our work starts.

The Blind Spot Paradox

The problem

Our reading, in one map

The blind spot is not a race between two clocks, but a budget of evidence set against a threshold.



Here is the whole talk on one map. At the top, the claim of the paper. The first column formalizes the race: we define the blind spot properly, we compute its probability, and we show that assuming independent trees makes the paper optimistic about the repair. The second column asks what the repair date really measures: it is the earliest possible date, it fires even when nothing changes, and when it fires most of the evidence is already there. The third column ties both together: the detector wins exactly when its evidence reaches the threshold before the repair, and that evidence explains both the transition and the paradox. The dashed arrows are the links between columns, and the box at the bottom is the question we leave open. Our one-sentence reading: the blind spot is not a race between two clocks, it is a budget of evidence set against a threshold.

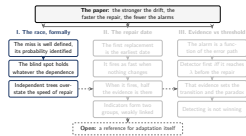
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The Blind Spot Paradox

I. The race

Part I — The race, formally

Let us start with the theory: how to state the race without cheating.



The Blind Spot Paradox

I. The race

— The race is well posed, and its probability identified

The paper says there is a blind spot when the forest repairs before the alarm, but it quietly sets aside the runs where the alarm never comes. Those are exactly the runs where the detector fails most. So we define the event on the complete race, with one definition and no extra rule. A tie counts for the detector, because the operator is warned at the right moment. A silent repair, where the alarm never comes, is a miss, and in fact the purest one. A run where neither mechanism acts is not a miss, because there was nothing to erase. Then comes the problem of the finite horizon: some runs could be undecided. On our campaign they are not, because the forest always acts within the horizon, so the miss probability is identified exactly: nine percent at threshold eight, ninety-seven at twenty-five, one hundred at fifty. And with Fréchet bounds, which use the two marginal laws and assume nothing on how the clocks are coupled, the blind spot at the weakest drift is already forced above ninety percent.

The race is well posed, and its probability identified

One definition, no ad hoc rule

$$\text{Miss} := \{\tau_{\text{ARP}} < \tau_{\text{Ae}}\}$$

strict inequality; the alarm may never come

- a tie goes to the detector
- a silent repair, with no alarm ever, is a miss
- if neither acts, it is not

What the campaign allows

The forest always replaces a tree within the horizon, so no run is left undecided: P_{blind} is identified.

	all 20 drift sizes pooled:		
λ	8	25	50
P_{blind}	0.0905	0.9715	1.0000

Without any dependence assumption

From the marginal laws alone, the blind spot at the weakest drift ($\Delta e = 0.028$, observed 1.00) is at least 0.91 at $\lambda = 8$.

τ_{ARP} : first tree replaced τ_{Ae} : first alarm Δe : drift size, the jump in error

The Blind Spot Paradox

└ I. The race

└ Trees share a stream: the paper is optimistic

Trees share a stream: the paper is optimistic

- Given the stream, what separates two trees is their own random draws: the trees are independent conditionally.
- Hence the covariance of two trees is a variance, never negative: sharing a stream can only make them fail together — the forest is slower than the independence formula says.
- More precisely, averaging over streams, convexity gives the bound:

$$P(\tau_{\text{FOREST}} \leq s) \leq 1 - (1 - F(s))^M$$
- Conclusion: the independence formula overstates the speed of the repair, hence the blind spot; the paper's critical ensemble size becomes a conservative ceiling, and its verdict does not change.

The second shortcut is independence. The ten trees learn from the same stream, so they cannot be independent. But the question is at which level. Once the stream is fixed, what distinguishes one tree from another is only its own randomness, the random weights of online bagging. At that level, given the stream, the trees are independent. That is a property of the algorithm, and everything else follows from it. First, the sign: the only channel between two trees is the stream they share, and a shared stream can only make them fail together, since their covariance reduces to a variance. So the real forest repairs more slowly than the independence formula says. Then the size: averaging over streams, Jensen's inequality gives a bound, the probability that the forest has already repaired itself is at most what the independence formula predicts. So the paper makes the forest look faster than it is, and overestimates the blind spot. Its critical ensemble size is not wrong, it becomes a conservative ceiling, and the paper's own verdict does not change.

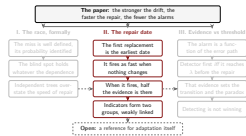
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The Blind Spot Paradox

II. The repair date

Part II — What the repair date measures

Now the second correction: the date the paper uses for the repair.



The Blind Spot Paradox

└ II. The repair date

└ How we measured

The campaign

- the paper's bench, rerun: 20 drift sizes $\times$ 100 seeds, 2000 steps after the change
- 2000 more runs with no change at all
- only the error and the replacement events are recorded; every threshold is replayed offline

Reading an association

- always within one drift size: pooling measures the drift size, not the link
- Kendall's rank correlation, with bootstrap intervals over the seeds
- steps added after seeing the data are flagged as exploratory

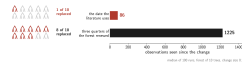
A word on how we measured, without the implementation details. We reran the bench of the paper on twenty drift sizes, with one hundred seeds each, and we observe two thousand steps after the change. We added two thousand runs where nothing changes at all, as a control. The simulation records only two things: the error at each step and the moments when trees are replaced. Everything else, including the detector at any threshold, is recomputed offline, so one campaign replaces many. When we correlate two quantities, we always do it within one drift size. The drift size moves every indicator at once, so a correlation pooled over sizes mostly measures the size itself. We use Kendall's rank correlation with bootstrap intervals, and we flag the steps we added after looking at the data.

The Blind Spot Paradox

II. The repair date

The first replacement is the earliest possible date

The first replacement is the earliest possible date



- On every run, the first replacement comes before any other quota of renewed trees. Three quarters of the forest are renewed 6.0 to 14.3 times later.
- With no drift at all, the first replacement comes after 95 steps in median, against 118 on the weakest drift.
- On weak drifts it comes later: 346 steps at $\Delta e = 0.085$. With no drift but the same error level, it also waits 268: the error level delays it, not the change.

The paper dates the repair at the first replaced tree. By construction, that is the earliest of all possible dates: the moment when three trees, or half, or three quarters of the forest are replaced, always comes later, on every run. The only question is how much later. On this medium drift, the first tree goes after eighty-six observations, and three quarters of the forest after more than twelve hundred. Over the grid, three quarters of the forest are renewed six to fourteen times later than the first tree. And there is a more worrying fact. On the two thousand runs with no change at all, the forest still replaces its first tree, after ninety-five steps in median. On the weakest real drift, it takes one hundred and eighteen. So at low drift, this date cannot tell a real change from noise. Worse, on a slightly stronger drift it comes later, after three hundred and forty-six steps. We checked why with a second control without drift, where we only raise the error level by flipping labels. At the same error level, the first replacement also waits about two hundred and seventy steps. The internal detector of each tree cuts less often when the error is noisier, so on weak drifts this date measures the error level, not the change.

The Blind Spot Paradox

└ II. The repair date

└ When the first tree goes, most of the evidence is already there

Is the first replacement at least the end of the transient? No. When the first tree is replaced, the detector already holds between half and three quarters of all the evidence it will ever accumulate. The replacement is in the middle of the story, not at its end. Then we asked whether this date tracks anything, always within a drift size. With the peak of the detector, there is a weak but real link, visible on fifteen drift sizes out of eighteen. With the excess error over the whole horizon, there is no link at all. But we found out why: that area adds up two thousand steps, so any error on the baseline is multiplied by two thousand, and half of its variance is just that noise. If we keep only the transition, the link comes back, weak but real, on thirteen sizes out of eighteen. This last step is exploratory.

When the first tree goes, most of the evidence is already there

Evidence at the first replacement

49% to 76%
of the detector's final peak, on
the 18 drift sizes studied

Does the date track anything, within a drift size?

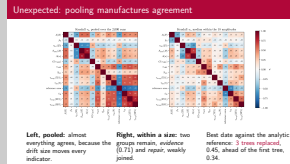
compared with	median link	sizes
detector peak	0.21	15/18
excess error, whole horizon	none	0/18
excess error, transition only	0.23	13/18

The null over the whole horizon is mostly noise: half of that area's variance comes from the estimated baseline, multiplied by the horizon.

The Blind Spot Paradox

II. The repair date

Unexpected: pooling manufactures agreement



To see all indicators at once, we crossed fourteen of them. The left panel pools all runs together: almost everything is strongly correlated with everything. That is an illusion, Simpson's paradox: the drift size moves every indicator, so pooling creates agreement. The right panel computes the same correlations within each drift size. Most of the agreement disappears, and two groups remain. One group measures evidence and damage: the detector's peak and the excess error over the transition agree strongly. The other group measures the repair. Between the two groups, links stay weak. We also compared the replacement dates with an analytic reference, the accuracy of the forest on the region whose label changed. The date at which three trees are replaced tracks it better than the first tree. This matrix is exploratory.

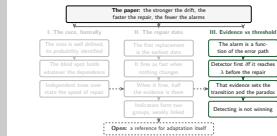
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The Blind Spot Paradox

III. Evidence vs threshold

Part III — Evidence against the threshold

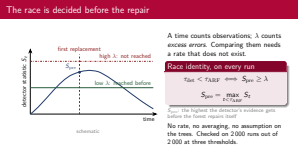
The third part ties the first two together.



The Blind Spot Paradox

III. Evidence vs threshold

The race is decided before the repair



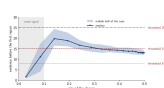
Here is the key step. A repair time counts observations. A threshold counts excess errors. They are not in the same unit, and the paper bridges them with an average rate of errors per step. That rate does not exist: it falls as the forest repairs, and the detector forgets its evidence each time it hits zero. The exact statement is much simpler. Look at the statistic of the detector up to the first replacement, and take its highest point: we call it S_{pre} . The detector arrives first if and only if S_{pre} reaches the threshold. With a low threshold, it is reached before the repair; with a high one, it is not. This is exact on every run, it needs no rate, no averaging and no assumption on the trees, and we checked it on all two thousand runs.

The Blind Spot Paradox

III. Evidence vs threshold

The evidence before the repair sets the transition

So we measured S pre on every run. The dark line is the median by drift size, the band holds the middle half of the runs, and the red lines are thresholds. Above the weak-signal zone, the median sits between thirteen and twenty. A threshold below that level is reached before the repair in most runs; a threshold above it rarely is. That is exactly the transition the paper observes between fifteen and twenty-five, and leaves to future work. Now look at the shape. After the medium drifts, S pre decreases as the drift gets stronger, because the first replacement comes sooner. At a fixed threshold of fifteen, the detector wins eighty-one percent of the races on medium drifts, and only eighteen percent on the strongest ones. The paradox is this decreasing evidence, read against a fixed threshold. At the far left, the drift is too weak to accumulate anything: that is a different regime.

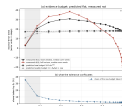


- Above the weak-signal zone, the median lies between 13.1 and 19.8: the transition the paper sees between 15 and 25, and leaves unexplained.
- It decreases as the drift grows: at $\lambda = 15$ the detector wins 81%, then only 18%. That is the paradox.

The Blind Spot Paradox

III. Evidence vs threshold

Stronger drifts leave no more evidence



- ♥ The paper's picture: an excess error Δe over a window τ_{ASF} , so the area stays almost constant.
- ♥ Measured over the transition, the area varies by a factor of 1.8 while the drift size varies by 5.8: the compensation is real.
- ♥ But the paper's rectangle misestimates it by up to 43%, and over the whole horizon the area turns negative at large drifts: the new task is easier.

Why doesn't a violent drift leave more evidence than a gentle one? The paper's picture is a rectangle: an excess error of height Δe , during a window equal to the repair time. Since the repair time shrinks like one over Δe , the area stays almost constant. We measured the area directly. Over the transition, it varies by a factor of less than two, while the drift size varies by almost six: the compensation is real. But the rectangle itself is wrong. It misestimates the area by up to forty-three percent, because the first replacement is not the width of the transient. And over the whole horizon, at large drifts, the area even becomes negative: after the change the task is easier than before, and the error settles below its old level. The evidence exists, it is roughly constant, but the paper's derivation of it does not hold.

The Blind Spot Paradox

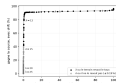
└ III. Evidence vs threshold

└ Detecting is not winning

One practical consequence. Detecting a drift and winning the race are two different things. At threshold fifteen, the alarm fires on ninety-three percent of drifts, which looks excellent, but it arrives before the repair in only thirty-six percent of them. Most of these alarms come after the forest has already changed. At threshold eight, the detector wins ninety-one percent of the races, and about four percent of the runs where nothing happens raise a false alarm within two thousand steps. At threshold twelve, it still wins seventy-nine percent of the races, for half a percent of false alarms. So on this bench, low thresholds do escape the blind spot, and the paper's claim that no threshold escapes has to be restricted. We did not test the financial streams with clustered volatility, where the paper places the false-alarm problem.

λ	alarm	first	false alarms
8	95.3%	81.0%	4.15%
10	94.8%	90.0%	1.30%
12	94.3%	78.8%	0.55%
15	92.9%	36.3%	0.15%
20	66.2%	8.1%	0.00%
25	38.9%	2.5%	0.00%

2000 runs with drift and 2000 without, over 2000 steps

each point is one λ , the time size just below $\lambda = 12$

- At $\lambda = 15$ the alarm almost always fires, but mostly after the repair.
- At $\lambda = 10$ the detector still wins 90.0% of races for 1.30% of false alarms: the paper's "no threshold escapes" does not hold on this bench.
- Not tested: the heteroscedastic streams where the paper places its false-alarm tension.

The Blind Spot Paradox

└ Conclusion

└ What changes in the paper

The paper	Our result	
race the repair against a constant	race identified, bounded, decided run by run	made exact
independent trees, bias asserted	direction proven, cannot reverse	proven
error back to normal at the first replacement	it comes back later, the rectangle fails	corrected
first replacement as the repair date	earliest date, fires without drift	qualified
transition between 15 and 25, unexplained	set by the evidence before the repair	explained
no threshold escapes	$\lambda = 8$ and 12 do, on this bench	restricted

The phenomenon survives. What changes is its mechanism and its reach.

To sum up what our work changes in the paper. The race, which compared a random time with a constant, is made exact: identified, bounded, and decided run by run. The bias of the independence assumption was asserted; it is now proven, with its direction. The idea that the error is back to normal at the first replacement is corrected. The first replacement as a repair date is qualified: it is the earliest date and it fires without drift. The transition threshold, which the paper left unexplained, is explained by the evidence before the repair. And the claim that no threshold escapes is restricted to the settings the paper actually tested. The phenomenon survives all of this. What changes is its mechanism, and its reach.

The Blind Spot Paradox

└ Conclusion

└ An open question, and what we did not do

Open: a reference for adaptation

Every indicator on the detector's side reads the error, which mixes what the forest relearned with how hard the new task is. On analytic probes, the forest can relearn the changed region and lose a rare one, unseen by the error.

Not done

- one forest size, one homoscedastic bench
- decision rule fixed during the analysis
- the independence bias has a sign, not a size

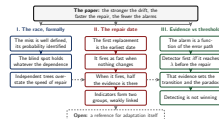
We leave one question open. Every measure available to the detector reads the error, and the error mixes two things: what the forest has relearned, and how hard the new task is. Using analytic probes, points whose true label we know, we saw that the forest can relearn the region that changed while losing a rare region elsewhere, and the error does not see that loss. So what would a true reference for adaptation be? We do not have the answer. And to be honest about the limits: we studied one forest size on one simple bench, our decision rule was fixed during the analysis rather than before it, and we can give the sign of the independence bias but not its size.

The Blind Spot Paradox

Conclusion

Take-away

The blind spot is a budget of evidence, accumulated before the repair, set against a threshold.



Back to the map. The blind spot is real, and it survives every correction. But it is not a race between two clocks. It is the evidence the forest lets through before its first repair, set against a fixed threshold. That single quantity explains when the detector wins, why the transition sits where it does, and why stronger drifts are missed more often. Thank you for your attention; we are happy to take your questions.